Weighted bounds for eliminating finite algebraic choices

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Abstract

We give a one-coordinate polynomial presentation of the rational points of a guarded finite free algebra with reduced active fibers. A direct scaling of its characteristic polynomials excludes singular choices of separating element and all drops in degree. Weighted principal minors yield explicit degree and coefficient bounds. Applied to the specified elliptic constructions, the method gives a family in twenty-seven total variables of degree less than 2^{151}, and a fully specified polynomial in twenty-nine total variables of degree less than 2^{157} and coefficient height at most $2^{2^{168}}$. Their assertions about defining the integers retain exactly the arithmetic hypotheses of their sources; the algebraic elimination theorem is unconditional.

1 The guarded finite algebra

Let $R = \mathbb{Z}[x_1, \dots, x_b]$, and let

$$A = R[z_1, \dots, z_m]/(f_1, \dots, f_m)$$

be a triangular monic presentation. Thus f_i involves only $z_1, \dots, z_i$ and is monic of degree e_i in z_i. The standard monomials

$$z_1^{a_1} \cdots z_m^{a_m}, \qquad 0 \le a_i < e_i,$$

are an R-basis of A. Write $n = \prod_i e_i$ for its rank and M_a for multiplication by $a \in A$ in this basis. Fix a guard $G \in A$. At every rational base point we assume that each geometric point of the fiber with $G \neq 0$ has local length one. No assumption is made about points where $G = 0$; those local algebras may be nonreduced. The predicate to be presented is

$$\text{there is an } R\text{-algebra specialization } A \longrightarrow \mathbb{Q} \text{ over the given base point with } G \neq 0. \quad (1)$$

There are no additional closed residual equations in this statement. The two applications below have precisely this form: every equation has already been included among the monic relations, and the remaining conditions are nonzero guards.

For $0 \le \ell \le mn(n+1)/2$, put

$$\lambda_\ell = 1 + \sum_{i=1}^{m} \ell^i z_i, \qquad H_\ell(V) = \det(I - V M_{G\lambda_\ell}) = \sum_{k=0}^{n} a_k V^k.$$

The coefficients $a_k \in R$ depend on ℓ, and $a_0 = 1$. For $1 \le d \le n$, define

$$H_{\ell,d}(V) = \sum_{k=0}^{d} a_k V^k, \qquad g_{\ell,d} = \operatorname{Res}_V(H_{\ell,d}, H'_{\ell,d}).$$

The resultant is the Sylvester determinant with prescribed degrees $d, d-1$, including after specialization. Define

$$Q_{\ell,d}(\mathbf{x}, U) = \begin{cases} H_{\ell,d}(g_{\ell,d}U)^2 + \sum_{k=d+1}^{n} a_k^2, & d < n, \\ H_{\ell,n}(g_{\ell,n}U), & d = n, \end{cases} \tag{2}$$

and set

$$P(\mathbf{x}, U) = \prod_{\ell=0}^{mn(n+1)/2} \prod_{d=1}^{n} Q_{\ell,d}(\mathbf{x}, U). \tag{3}$$

All these expressions are integer polynomials. Their indices, matrices and algebra generators are part of a finite coefficient specification, not additional existential coordinates.

Theorem 1.1. *For every rational base point* $\mathbf{x}$*, there is a rational* U *with* $P(\mathbf{x}, U) = 0$ *if and only if* (1) *holds. In particular, the construction adds exactly one rational coordinate.*

Proof. The universal identity

$$\operatorname{Res}(f, f') = (-1)^{d(d-1)/2} a_d \operatorname{Disc}(f)$$

shows that $g_{\ell,d} \neq 0$ forces $a_d \neq 0$ and makes $H_{\ell,d}$ squarefree of degree d. This also holds for $d = 1$, when the discriminant is one.

Suppose $Q_{\ell,d}(\mathbf{x}, U) = 0$. If $g_{\ell,d} = 0$ or $U = 0$, then $H_{\ell,d}(g_{\ell,d}U) = 1$, a contradiction. For $d < n$, a sum of rational squares vanishes only when all its terms vanish. Thus $a_{d+1} = \cdots = a_n = 0$ and $H_\ell = H_{\ell,d}$. The same equality is automatic for $d = n$. Consequently $V = g_{\ell,d}U$ is a nonzero simple rational root of H_ℓ. With $M = M_{G\lambda_\ell}$, the rational matrix $I - VM$ has a one-dimensional kernel. Every multiplication matrix M_{z_i} preserves this kernel and acts there by a rational scalar ζ_i. The algebra relations imply $f_i(\boldsymbol{\zeta}) = 0$. The eigenvalue of M there is $1/V \neq 0$, so $G(\boldsymbol{\zeta}) \neq 0$. This proves soundness, including fibers with inactive nilpotents and complex residue fields.

Conversely, suppose a guarded rational point exists. At each geometric point p with $G(p) \neq 0$, the polynomial in ℓ

$$G(p)\left(1 + \sum_i \ell^i z_i(p)\right)$$

is nonzero and has degree at most m. For distinct such points p, q, the difference of the corresponding polynomials is nonzero: otherwise their constant coefficients give $G(p) = G(q) \neq 0$, and their other coefficients give $z_i(p) = z_i(q)$ for every i. There are at most n points. Counting zero values and pairwise collisions, at most $mn(n+1)/2$ values of ℓ are excluded. Hence one of the indicated integers makes all active eigenvalues nonzero and distinct.

At an inactive local algebra multiplication by $G\lambda_\ell$ has only the eigenvalue zero, so its contribution to $\det(I - VM)$ is one, even if the algebra is nonreduced. All active local algebras have length one. Therefore, for the selected ℓ, H_ℓ is squarefree of some degree $1 \leq d \leq n$. The given rational point supplies a nonzero rational root V. Its resultant $g_{\ell,d}$ is nonzero. The rational choice $U = V/g_{\ell,d}$ annihilates the corresponding factor in (2) and hence the product. □

The proof permits degree drops and does not require the algebra or its inactive fibers to be totally real. It does not replace a branchwise nonzero guard by the stronger requirement that its norm be nonzero on every geometric branch.

2 Weighted principal minors

Give each base coordinate weight one and each z_i a nonnegative integer weight w_i. Assume that every nonleading monomial in f_i has weight at most $e_i w_i$. Monic reduction then never increases weight. Suppose $G\lambda_\ell$ has weight at most w, uniformly in ℓ.

Lemma 2.1. *For the characteristic coefficients above,* $\deg a_k \le kw$.

Proof. Let b_j be the weight of the jth standard basis monomial. The (i,j) entry of $M_{G\lambda_\ell}$ has degree at most $w + b_j - b_i$; a negative bound means that the entry is zero. For a principal minor indexed by a set I of size k, every nonzero determinant term has degree at most

$$\sum_{j\in I}(w + b_j - b_{\sigma(j)}) = kw.$$

The basis weights cancel. Since a_k, up to sign, is the sum of all principal minors of size k, the result follows. □

The resultant $\operatorname{Res}(H_d, H_d')$ is weighted homogeneous of weight d^2 when a_k has weight k. Indeed, replacing a_k by $s^k a_k$ replaces $H_d(V)$ by $H_d(sV)$, and the resultant by $s^{d^2}\operatorname{Res}(H_d, H_d')$. Hence

$$\deg g_{\ell,d} \le d^2 w, \qquad \deg H_{\ell,d}(g_{\ell,d}U) \le (d^3+d)w + d.$$

The degree cases can be summed separately. Write

$$S = 1 + \frac{mn(n+1)}{2}, \qquad C(n) = 2\sum_{d=1}^{n-1}\max(d^3+d, n) + (n^3+n).$$

Theorem 2.2. *The polynomial* (3) *satisfies*

$$\deg P \le S\big(wC(n) + n^2\big). \tag{4}$$

Proof. For $d < n$, the degree of its factor is at most

$$2\max\big((d^3+d)w + d, nw\big) \le 2w\max(d^3+d, n) + 2d.$$

For $d = n$ the unsquared factor has degree at most $(n^3+n)w + n$. The sum of the additive terms is $2\sum_{d=1}^{n-1} d + n = n^2$. There are S separating choices. □

If t is the largest integer with $t^3 + t < n$, then an explicit closed formula for the finite sum is

$$C(n) = \frac{n^4+3n^2}{2} + 2tn - \frac{t^2(t+1)^2}{2} - t(t+1). \tag{5}$$

3 Coefficient norms

For an integer polynomial F, write $\|F\|_1$ for the sum of the absolute values of its coefficients. Its coefficient height is at most $\|F\|_1$. Choose positive real weights for the standard basis and use the resulting weighted column norm on multiplication matrices, applying $\|\cdot\|_1$ to each polynomial entry. Suppose

$$\|M_{G\lambda_\ell}\| \le 2^\beta$$

uniformly in ℓ. Principal minors are invariant under the diagonal similarity defining this norm. Expanding a minor and using column sums therefore gives

$$\|a_k\|_1 \le \binom{n}{k} 2^{k\beta} \le 2^{k(\beta + \log_2 n)}.$$

In this section suppose $n = 2^r$, and put $b = \beta + r$.

Lemma 3.1. *For $1 \le d \le n$,*

$$\log_2\|g_{\ell,d}\|_1 \le d^2 b + 3d(r+1),$$

where the inequality is automatic if the resultant is zero.

Proof. Every nonzero term in the Sylvester determinant has coefficient index weight d^2, so the characteristic-coefficient contribution is at most $2^{d^2 b}$. There are d derivative rows, contributing at most d^d, and at most $(2d-1)!$ determinant terms. Their product is at most $(2d)^{3d}$, and $\log_2(2d) \le r+1$. □

It follows that

$$\log_2\|H_{\ell,d}(g_{\ell,d}U)\|_1 \le (d^3+d)b + 3d^2(r+1) + \lceil \log_2(d+1)\rceil.$$

For $d < n$, bounding the sum in (2) by the number of its summands gives

$$\log_2\|Q_{\ell,d}\|_1 \le 2b\max(d^3+d, n) + (6d^2+3)(r+1).$$

For $d = n$ the unsquared factor instead satisfies

$$\log_2\|Q_{\ell,n}\|_1 \le (n^3+n)b + (3n^2+1)(r+1).$$

Submultiplicativity of the coefficient norm proves the following bound.

Theorem 3.2. *Under the preceding hypotheses,*

$$\log_2\|P\|_1 \le S\left((\beta+r)C(n) + (r+1)(2n^3+4n-2)\right). \tag{6}$$

Proof. Sum the preceding inequalities over d and use

$$6\sum_{d=1}^{n-1} d^2 + 3(n-1) + 3n^2 + 1 = 2n^3 + 4n - 2.$$

Then sum over the S values of ℓ. □

4 The defining-polynomial applications

The arithmetic constructions in [1, 2] provide triangular monic algebras with no closed residual equations. The source data specify every coefficient and every nonzero guard. They retain respectively 26 and 28 base coordinates, including the distinguished input. Nine monic quartic relations describe the point coordinates. There are respectively three and four additional quadratic relations for pole witnesses. The point derivatives are nonzero at every geometric point over a rational base point; the quadratic roots themselves are guarded nonzero. Thus the active local-length hypothesis of Theorem 1.1 holds.

The following values are obtained by integer recurrences directly from the monic relation coefficients and the guard.

Quantity	Twenty-seven variables	Twenty-nine variables
m	12	13
n	2^{21}	2^{22}
w	6 358 790	8 724 856
β at $L = 137731$	12 880 956 901	12 887 537 402
t in (5)	127	161

For completeness, the generator weights for the first algebra are

$$\underbrace{6,\dots,6}_{9\text{ entries}},1568138,\ 1935231,\ 116318,$$

and the second has one additional weight 1935231. The guard weights are 4423559 and 6789625, respectively; adding the largest generator weight gives w. The coefficient norm bounds use the weighted companion-column recurrences of the specified monic algebras. In each relation, choose a logarithmic generator bound at least

$$\max_{k<e_i}\left\lceil\frac{\log_2\|c_{ik}\|}{e_i-k}\right\rceil+\lceil\log_2\#\{k:c_{ik}\neq 0\}\rceil,$$

where $f_i=z_i^{e_i}+\sum_{k<e_i}c_{ik}z_i^k$ and the previously chosen generator weights are used in the coefficient norm. This bounds multiplication by z_i. Applying addition and multiplication inequalities to the guard and separating element gives the displayed β uniformly over its finite selector range.

Corollary 4.1. *Under the arithmetic hypotheses of [1], there is a computable family in* 27 *total variables such that every sufficiently large odd fixed coefficient exponent defines* $\mathbb{Z}$ *over* $\mathbb{Q}$ *and*

$$\deg F\le 162283885779556443896478012052741319971315788 8<2^{151}.$$

The sufficient coefficient threshold remains ineffective.

Proof. Apply Theorem 1.1 to the 26-coordinate guarded source. Its rational base projection is preserved exactly. The source arithmetic theorem therefore applies with the same hypotheses and coefficient threshold. The source weights are uniform in the odd exponent because that exponent occurs only in fixed coefficients. Substitution into (4) gives the bound. □

Corollary 4.2. *Under the arithmetic hypotheses of [2], the same construction applied to its specified source with fixed exponent* $L=137731$ *gives an integer polynomial in* 29 *total variables which defines* $\mathbb{Z}$ *over* $\mathbb{Q}$*, with*

$$\deg F\le 154383598645603405051555717167316362269845989152<2^{157},$$

and coefficient height at most 2^{C_*}*, where*

$$C_*=228040944767112830511621792043639838233373391751570<2^{168}.$$

Proof. Theorem 1.1 preserves the rational base projection of the 28-coordinate source. Its fixed parameter is the certified sufficient one from the effective arithmetic theorem. Substitute the second column of data into (4) and (6). □

Both degree assertions are upper bounds for completely specified polynomials, not claims that the expanded polynomials attain those degrees. The first corollary does not certify 137731 as a sufficient parameter. The construction at that exponent is an algebraically specified example in the family; its computable coefficient bound cannot supply an effective bound for a member known to define $\mathbb{Z}$. Neither corollary removes an arithmetic premise or proves a smaller number of variables.

5 Reproducible certificates

The accompanying program `build_weighted_specification.py` reads a verified source specification, checks the absence of closed residuals and indexed guards, reconstructs every generator bound, and writes (3) as a complete indexed polynomial specification. The files for the 27- and 29-variable constructions include integer degree and coefficient certificates and SHA-256 identifiers for both their input and output specifications.

The independent program `check_weighted_specification.py` evaluates the defining sum $C(n)$ literally and checks the integer bounds. It also verifies small exact finite algebras with irrational real roots, complex residue factors, zero guards, inactive nilpotents, selector collisions, and degree drops. In every successful branch it recovers the rational character from the one-dimensional kernel. These tests exercise the exceptional cases in the proof; they do not replace that proof or verify the global arithmetic hypotheses.

References

[1] *A defining polynomial in twenty-seven variables: an existence theorem with a uniform degree bound*, companion paper; its exact guarded source is supplied in `computation/few_variables/trace27_nontorsion`.

[2] *An explicit defining polynomial in twenty-nine variables with effective elliptic sum heights*, companion paper; its exact guarded source is supplied in `computation/few_variables/effective29_nontorsion`.

[3] *Eliminating finite algebraic choices with one coordinate*, companion paper on general guarded finite algebras, including closed residual equations and arbitrary number fields.